to determine the concentration of Li_3N following a salicylate assay[36]. The water titration scheme is depicted in Supplementary Fig. 7.

For 2-butoxyethanol titration, special care was taken to further suppress contamination of the materials used in the analysis by water. As such, after regular vacuum drying at 70 °C, all materials and utensils, including vials, tweezers, syringes and needles, were dried in a vacuum glass oven at 70 °C for at least 12 hours and taken inside of the glovebox without any atmospheric exposure. Inside of the glovebox, the cycled Cu current collector and the separator closest to it were carefully extracted from the cell and closed inside a dry gas-tight vial without rinsing. Immediately after, 500 µl of dry butoxyethanol (typically ~3 ppm H_2O) was injected into the vial through a septum using a dry gas-tight syringe. After ~24 hours, when the sample had fully reacted, another 1.5 ml of butoxyethanol was added to the vial, resulting in a 2 ml solution per sample. The solution was then analysed by KF titration to determine the concentration of Li_2O in the solution. The 2-butoxyethanol titration scheme is depicted in Supplementary Fig. 8.

Coulometric KF titration of Li_2O

Upon reaction of the Li residuals with 2-butoxyethanol, Li_2O was converted into LiOH following the reaction $ROH + Li_2O \rightarrow LiOH + ROLi$ ($R = CH_3(CH_2)_3O(CH_2)_2$ for 2-butoxyethanol). The KF electrolyte (CombiCoulomat fritless) is methanol-based and contains a small amount of hydroiodic acid (HI) due to the presence of iodine (I_2) in the solution, which is needed for KF titration. When added to the electrolyte, the hydroxide in the sample solution is consumed by HI, forming water stoichiometrically ($LiOH + HI \rightarrow H_2O + LiI$), which then follows the typical Karl Fischer reaction by consuming iodine ($H_2O + I_2 + SO_2 + 3\,R_bN + CH_3OH \rightarrow 2\,R_bN\,HI + RN\,HSO_4CH_3$, where R_bN is a buffering agent, typically pyridine). Iodine is then rebalanced coulometrically by the generator electrode in the KF titrator, and the charge needed for rebalancing is recorded ($2\,I^+ + 2\,e^- \rightarrow I_2$). Hence, there is a direct and stoichiometric relationship between the original amount of Li_2O in the sample and the charge measured for rebalancing iodine ($2\,e^-_{KF} = 1\,Li_2O$), as shown in Fig. 1b. More details on KF titration are in the Supplementary Methods.

Cryogenic transmission electron microscopy

Cu/Li cells used for Cryo-TEM were assembled as usual, with the only difference being the addition of three TEM Cu grids (Ted Pella, 1GC300) directly on top of the Cu current collector. Li was directly plated on the TEM grids at 0.5 mA cm^{-2} to a capacity of either 0.1 or 0.2 mAh cm^{-2} and immediately taken inside of the glovebox and disassembled for analysis. After disassembly, the grids were carefully rinsed by droplets of DMC or DME based on the solvent of the electrolyte, and subsequently grids were placed on a piece of Kimwipe in the glovebox for ten minutes to dry out the remaining rinsing solvents, then grids were sealed in Eppendorf tubes separately with Parafilm to ensure air-tight transfer. Afterwards, the Eppendorf tubes with grids inside were transferred outside of the glovebox, immediately immersed in a Styrofoam container filled with liquid nitrogen, and the Eppendorf tubes were crushed by a pair of pliers to immerse the grids in the liquid nitrogen (LN2). Grids were transferred into a grid box within LN2 and then stored in a Dewar for following cryo analysis.

Cryo analysis was conducted by an instrument in the Electron Imaging Center for Nanomachines at California NanoSystems Institute, using a FEI Titan 80–300 scanning transmission electron microscope operated at an accelerating voltage of 300 kV, which is equipped with a field emission gun, an Oxford X-MaxTEM 100 N TLE windowless SDD 100 mm^2 for EDS and Ultrascan US1000 1 K digital camera. The grids were transferred into the TEM column with Gatan 626 cryo-transfer holder and the grids were immersed in LN2 during the whole transfer process to avoid potential air contamination. After insertion of the cryo holder, the temperature was maintained at −178 °C. The electron flux under low magnification was around 100 e $Å^{-2}$ s^{-1} and 1,000 e $Å^{-2}$ s^{-1} under high magnification. The acquisition time for SAED images was 0.05 s and for TEM images was 0.4 to 2 s.

Data availability

The data supporting the findings of this study are included within the article and its Supplementary Information files. Source data are provided with this paper.

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Author contributions

G.M.H. and B.M.G. conceived the idea and designed the experiments. G.M.H. carried out experiments and measurements. C.W. and Y.L. co-designed Cryo-HRTEM experiments and performed imaging. K.S. assisted with method development. G.M.H. and B.M.G. analysed the data and prepared the initial draft of the paper. All authors discussed the results and contributed to the final version of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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